MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE 5SS

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductor, tin coated, AWG No. 18 (41/No.34).
Second - Separator.
Third - Synthetic rubber insulation. 0.032 inch nominal thickness. Identification by Method 3; one conductor Black, 2-White, 1-Red and 1-Green.
Fourth - Braided shield (one white conductor only) of AWG No. 34 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 88 percent.
Fifth - Shield insulation of one 0.002 inch thick polyester tape applied helically with 50 percent minimum overlap.
Sixth - The five conductors cabled with a 3.75 ± 0.25-inch right hand lay around a moisture resistant jute center. Cabling sequence shall be black, white, white with shield, green and red.
Seventh - Binder tape of 0.012 inch thick double-faced, rubber-filled cotton tape applied helically with overlap.
Eighth - Overall braided shield of AWG No. 34 tin-coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 88 percent.
Ninth - Shield insulation of one 0.002 inch thick polyester tape applied helically with 50 percent overlap.
Tenth - Polychloroprene jacket. Minimum thickness 0.050 inch. Cable diameter 0.500 inch minimum; 0.560 inch maximum. Surface of cable shall be embossed at approximately 3-foot intervals with the following information: The year of manufacture, Type 5SS 5 CONDR No. 18 SHLD 600 VOLTS.

EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - Ohms/1000 ft. at 25°C. maximum	7.05
Voltage withstand - Volts, rms. minimum	
Conductor to conductor	3000
Conductor to shield	1000
Insulation resistance - Megohms/1000 ft., minimum	10,000
Group A:	
Visual and dimensional.	
Group B:	
Physicals (unaged)	
Insulation	
Tensile strength - Psi, minimum	1000
Elongation - Percent, minimum	500
Set-inch, maximum	3/8
Jacket	
Tensile strength - Psi, minimum	2400
Elongation - Percent, minimum	325
Tensile stress - Psi, minimum at 200 percent elongation (ASTM test method D470)	700
Set - Inches, maximum	1/4
Tear - Pounds/inch thickness, minimum (ASTM test method D470)	40
Bending endurance - Cycles, minimum at minus 30° + 2°C. on 4-inch maximum diameter mandrel.	1000

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:16Z
Check the source to verify that this is the current version before use.

	Requirements:
Group C:	
Physicals (aged).	
Insulation	
Air pressure heat.	
Tensile strength - Percent of unaged, minimum	75
Elongation - Percent of unaged, minimum	60
Oxygen pressure.	
Tensile strength - Percent of unaged, minimum	75
Elongation - Percent of unaged, minimum	60
Jacket	
Air pressure heat.	
Tensile strength - Percent of unaged, minimum	65
Elongation - Percent of unaged, minimum	60
Oxygen pressure.	
Tensile strength - Percent of unaged, minimum	65
Elongation - Percent of unaged, minimum	60

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C.

UNIT ORDERING LENGTHS:

2000-foot continuous lengths, except that 20 percent of total quantity may be in lengths between 1000 and 2000 feet but none less than 1000 feet.

NOTE: Cable shall be shipped on non-returnable reels, one length per reel and the outside end of cable shall have the conductors in clock-wise sequence as follows: Green, Red, Black, White and White-Shielded.

Preparing activity:
Navy - SH
(Project 6145-N227)

☆ U. S. GOVERNMENT PRINTING OFFICE: 1972 -714-539/2380

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:16Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:16Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commander, Naval Ship Engineering Center
Code 6124, Center Building
Prince George's Center
Hyattsville, Maryland 20782

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:16Z
Check the source to verify that this is the current version before use.